

ĐẠI HỌC QUỐC GIA THÀNH PHỐ HỒ CHÍ MINH
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ THÔNG TIN
KHOA CÔNG NGHỆ PHẦN MỀM



MÔN LẬP TRÌNH TRỰC QUAN
BÀI TẬP THỰC HÀNH LAB 05

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This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

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(Ký tên và ghi rõ họ tên)

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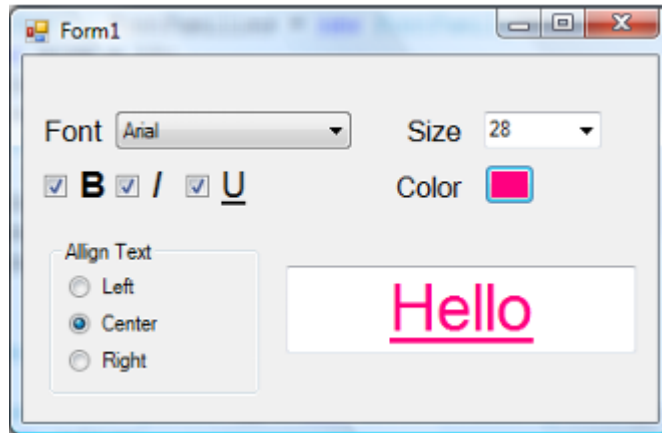
DANH MỤC HÌNH ẢNH

Hình 1. 1 Class diagram của lớp NgayThangNam.....**Error! Bookmark not defined.**

Hình 1. 2 xây dựng lớp, thuộc tính, phương thức.**Error! Bookmark not defined.**

NỘI DUNG BÀI LÀM

Bài 04. Xây dựng ứng dụng xử lý font văn bản như sau:



- Cho chọn kiểu Font (gồm các font có sẵn trong hệ thống).
- Chọn kích thước (Size), in đậm (B), in nghiêng (I), gạch dưới (U).
- Canh lề trái, phải hoặc giữa.
- Chọn màu chữ: khi nhấn button, hiện ColorDialog cho người dùng chọn màu.

Bảng 1-1 Nội dung code Form1.Designer.cs của Bài 04

```
namespace Bai04
{
    partial class Form1
    {
        /// <summary>
        /// Required designer variable.
        /// </summary>
        private System.ComponentModel.IContainer components = null;

        /// <summary>
        /// Clean up any resources being used.
        /// </summary>
        /// <param name="disposing">true if managed resources should be disposed;
        otherwise, false.</param>
        protected override void Dispose(bool disposing)
        {
            if (disposing && (components != null))
            {
                components.Dispose();
            }
            base.Dispose(disposing);
        }

        #region Windows Form Designer generated code

        /// <summary>
        /// Required method for Designer support - do not modify
        /// the contents of this method with the code editor.
    }
}
```

```

/// </summary>
private void InitializeComponent()
{
    this.label1 = new System.Windows.Forms.Label();
    this.cbbFont = new System.Windows.Forms.ComboBox();
    this.cbbSize = new System.Windows.Forms.ComboBox();
    this.label2 = new System.Windows.Forms.Label();
    this.btnColor = new System.Windows.Forms.Button();
    this.label3 = new System.Windows.Forms.Label();
    this.chbUnderline = new System.Windows.Forms.CheckBox();
    this.chbItalic = new System.Windows.Forms.CheckBox();
    this.chbBold = new System.Windows.Forms.CheckBox();
    this.grbAlignText = new System.Windows.Forms.GroupBox();
    this.radRight = new System.Windows.Forms.RadioButton();
    this.radCenter = new System.Windows.Forms.RadioButton();
    this.radLeft = new System.Windows.Forms.RadioButton();
    this.panel1 = new System.Windows.Forms.Panel();
    this.txbText = new System.Windows.Forms.TextBox();
    this.grbAlignText.SuspendLayout();
    this.panel1.SuspendLayout();
    this.SuspendLayout();
    //
    // label1
    //
    this.label1.AutoSize = true;
    this.label1.Font = new System.Drawing.Font("Microsoft Sans Serif",
13.8F, System.Drawing.FontStyle.Bold, System.Drawing.GraphicsUnit.Point,
((byte)0));
    this.label1.Location = new System.Drawing.Point(22, 42);
    this.label1.Name = "label1";
    this.label1.Size = new System.Drawing.Size(72, 29);
    this.label1.TabIndex = 0;
    this.label1.Text = "Font:";
    //
    // cbbFont
    //
    this.cbbFont.BackColor =
System.Drawing.Color.FromArgb(((int)(((byte)224))), ((int)(((byte)224))),
((int)(((byte)224))));
    this.cbbFont.DropDownStyle =
System.Windows.Forms.ComboBoxStyle.DropDownList;
    this.cbbFont.FormattingEnabled = true;
    this.cbbFont.ImeMode = System.Windows.Forms.ImeMode.NoControl;
    this.cbbFont.Location = new System.Drawing.Point(100, 47);
    this.cbbFont.Name = "cbbFont";
    this.cbbFont.Size = new System.Drawing.Size(195, 24);
    this.cbbFont.TabIndex = 1;
    this.cbbFont.SelectedIndexChanged += new
System.EventHandler(this.FormatChanged);
    //
    // cbbSize
    //
    this.cbbSize.BackColor =
System.Drawing.Color.FromArgb(((int)(((byte)224))), ((int)(((byte)224))),
((int)(((byte)224))));
    this.cbbSize.DropDownStyle =
System.Windows.Forms.ComboBoxStyle.DropDownList;
    this.cbbSize.FormattingEnabled = true;
    this.cbbSize.ImeMode = System.Windows.Forms.ImeMode.NoControl;
    this.cbbSize.Items.AddRange(new object[] {
        "8",
        "9",
        "10",
        "11",
        "12",
        "14",
        "16",
        "18",
        "20",

```

```

        "22",
        "24",
        "26",
        "28",
        "36",
        "48",
        "72"});
this.cbbSize.Location = new System.Drawing.Point(442, 49);
this.cbbSize.Name = "cbbSize";
this.cbbSize.Size = new System.Drawing.Size(83, 24);
this.cbbSize.TabIndex = 3;
this.cbbSize.SelectedIndexChanged += new
System.EventHandler(this.FormatChanged);
//
// label2
//
this.label2.AutoSize = true;
this.label2.Font = new System.Drawing.Font("Microsoft Sans Serif",
13.8F, System.Drawing.FontStyle.Bold, System.Drawing.GraphicsUnit.Point,
((byte)0));
this.label2.Location = new System.Drawing.Point(364, 44);
this.label2.Name = "label2";
this.label2.Size = new System.Drawing.Size(71, 29);
this.label2.TabIndex = 2;
this.label2.Text = "Size:";
//
// btnColor
//
this.btnColor.Location = new System.Drawing.Point(442, 106);
this.btnColor.Name = "btnColor";
this.btnColor.Size = new System.Drawing.Size(41, 37);
this.btnColor.TabIndex = 4;
this.btnColor.UseVisualStyleBackColor = true;
this.btnColor.Click += new System.EventHandler(this.btnColor_Click);
//
// label3
//
this.label3.AutoSize = true;
this.label3.Font = new System.Drawing.Font("Microsoft Sans Serif",
13.8F, System.Drawing.FontStyle.Bold, System.Drawing.GraphicsUnit.Point,
((byte)0));
this.label3.Location = new System.Drawing.Point(351, 110);
this.label3.Name = "label3";
this.label3.Size = new System.Drawing.Size(84, 29);
this.label3.TabIndex = 5;
this.label3.Text = "Color:";
//
// chbUnderline
//
this.chbUnderline.AutoSize = true;
this.chbUnderline.Image =
global::Bai04.Properties.Resources.icons8_underline_30;
this.chbUnderline.Location = new System.Drawing.Point(247, 109);
this.chbUnderline.Name = "chbUnderline";
this.chbUnderline.Size = new System.Drawing.Size(48, 30);
this.chbUnderline.TabIndex = 8;
this.chbUnderline.UseVisualStyleBackColor = true;
this.chbUnderline.CheckedChanged += new
System.EventHandler(this.FormatChanged);
//
// chbItalic
//
this.chbItalic.AutoSize = true;
this.chbItalic.Image =
global::Bai04.Properties.Resources.icons8_italic_30;
this.chbItalic.Location = new System.Drawing.Point(182, 109);
this.chbItalic.Name = "chbItalic";
this.chbItalic.Size = new System.Drawing.Size(48, 30);
this.chbItalic.TabIndex = 7;

```

```

        this.chbItalic.UseVisualStyleBackColor = true;
        this.chbItalic.CheckedChanged += new
System.EventHandler(this.FormatChanged);
        //
        // chbBold
        //
        this.chbBold.AutoSize = true;
        this.chbBold.Image = global::Bai04.Properties.Resources.icons8_bold_30;
        this.chbBold.Location = new System.Drawing.Point(109, 109);
        this.chbBold.Name = "chbBold";
        this.chbBold.Size = new System.Drawing.Size(48, 30);
        this.chbBold.TabIndex = 6;
        this.chbBold.UseVisualStyleBackColor = true;
        this.chbBold.CheckedChanged += new
System.EventHandler(this.FormatChanged);
        //
        // grbAlignText
        //
        this.grbAlignText.BackColor = System.Drawing.SystemColors.ButtonFace;
        this.grbAlignText.Controls.Add(this.radRight);
        this.grbAlignText.Controls.Add(this.radCenter);
        this.grbAlignText.Controls.Add(this.radLeft);
        this.grbAlignText.Font = new System.Drawing.Font("Microsoft Sans
Serif", 12F, System.Drawing.FontStyle.Regular, System.Drawing.GraphicsUnit.Point,
((byte)(0)));
        this.grbAlignText.Location = new System.Drawing.Point(22, 223);
        this.grbAlignText.Name = "grbAlignText";
        this.grbAlignText.Size = new System.Drawing.Size(128, 204);
        this.grbAlignText.TabIndex = 9;
        this.grbAlignText.TabStop = false;
        this.grbAlignText.Text = "Align Text";
        //
        // radRight
        //
        this.radRight.AutoSize = true;
        this.radRight.Font = new System.Drawing.Font("Microsoft Sans Serif",
12F, System.Drawing.FontStyle.Regular, System.Drawing.GraphicsUnit.Point,
((byte)(0)));
        this.radRight.Location = new System.Drawing.Point(6, 131);
        this.radRight.Name = "radRight";
        this.radRight.Size = new System.Drawing.Size(77, 29);
        this.radRight.TabIndex = 2;
        this.radRight.TabStop = true;
        this.radRight.Text = "Right";
        this.radRight.UseVisualStyleBackColor = true;
        this.radRight.CheckedChanged += new
System.EventHandler(this.AlignChanged);
        //
        // radCenter
        //
        this.radCenter.AutoSize = true;
        this.radCenter.Font = new System.Drawing.Font("Microsoft Sans Serif",
12F, System.Drawing.FontStyle.Regular, System.Drawing.GraphicsUnit.Point,
((byte)(0)));
        this.radCenter.Location = new System.Drawing.Point(6, 87);
        this.radCenter.Name = "radCenter";
        this.radCenter.Size = new System.Drawing.Size(92, 29);
        this.radCenter.TabIndex = 1;
        this.radCenter.TabStop = true;
        this.radCenter.Text = "Center";
        this.radCenter.UseVisualStyleBackColor = true;
        this.radCenter.CheckedChanged += new
System.EventHandler(this.AlignChanged);
        //
        // radLeft
        //
        this.radLeft.AutoSize = true;

```

```

        this.radLeft.Font = new System.Drawing.Font("Microsoft Sans Serif",
12F, System.Drawing.FontStyle.Regular, System.Drawing.GraphicsUnit.Point,
((byte)(0)));
        this.radLeft.Location = new System.Drawing.Point(6, 43);
        this.radLeft.Name = "radLeft";
        this.radLeft.Size = new System.Drawing.Size(65, 29);
        this.radLeft.TabIndex = 0;
        this.radLeft.TabStop = true;
        this.radLeft.Text = "Left";
        this.radLeft.UseVisualStyleBackColor = true;
        this.radLeft.CheckedChanged += new
System.EventHandler(this.AlignChanged);
        //
        // panel1
        //
        this.panel1.Anchor =
((System.Windows.Forms.AnchorStyles)((((System.Windows.Forms.AnchorStyles.Top |
System.Windows.Forms.AnchorStyles.Bottom)
| System.Windows.Forms.AnchorStyles.Left)
| System.Windows.Forms.AnchorStyles.Right)));
        this.panel1.Controls.Add(this.chbUnderline);
        this.panel1.Controls.Add(this.chbItalic);
        this.panel1.Controls.Add(this.chbBold);
        this.panel1.Controls.Add(this.label3);
        this.panel1.Controls.Add(this.btnColor);
        this.panel1.Controls.Add(this.cbbSize);
        this.panel1.Controls.Add(this.label2);
        this.panel1.Controls.Add(this.cbbFont);
        this.panel1.Controls.Add(this.label1);
        this.panel1.Location = new System.Drawing.Point(22, 33);
        this.panel1.Name = "panel1";
        this.panel1.Size = new System.Drawing.Size(560, 170);
        this.panel1.TabIndex = 10;
        //
        // txtbText
        //
        this.txtbText.BackColor = System.Drawing.SystemColors.ButtonHighlight;
        this.txtbText.BorderStyle =
System.Windows.Forms.BorderStyle.FixedSingle;
        this.txtbText.Location = new System.Drawing.Point(156, 234);
        this.txtbText.Multiline = true;
        this.txtbText.Name = "txtbText";
        this.txtbText.ReadOnly = true;
        this.txtbText.Size = new System.Drawing.Size(425, 204);
        this.txtbText.TabIndex = 11;
        this.txtbText.Text = "Hello";
        //
        // Form1
        //
        this.AutoScaleMode = new System.Drawing.SizeF(8F, 16F);
        this.AutoScaleMode = System.Windows.Forms.AutoScaleMode.Font;
        this.ClientSize = new System.Drawing.Size(600, 450);
        this.Controls.Add(this.txtbText);
        this.Controls.Add(this.panel1);
        this.Controls.Add(this.grbAlignText);
        this.Name = "Form1";
        this.Text = "Bai 04 - Xử lý font";
        this.grbAlignText.ResumeLayout(false);
        this.grbAlignText.PerformLayout();
        this.panel1.ResumeLayout(false);
        this.panel1.PerformLayout();
        this.ResumeLayout(false);
        this.PerformLayout();

    }

#endregion

private System.Windows.Forms.Label label1;

```

```

private System.Windows.Forms.ComboBox cbbFont;
private System.Windows.Forms.ComboBox cbbSize;
private System.Windows.Forms.Label label2;
private System.Windows.Forms.Button btnColor;
private System.Windows.Forms.Label label3;
private System.Windows.Forms.CheckBox chbBold;
private System.Windows.Forms.CheckBox chbItalic;
private System.Windows.Forms.CheckBox chbUnderline;
private System.Windows.Forms.GroupBox grbAlignText;
private System.Windows.Forms.Panel panel1;
private System.Windows.Forms.TextBox txtText;
private System.Windows.Forms.RadioButton radRight;
private System.Windows.Forms.RadioButton radCenter;
private System.Windows.Forms.RadioButton radLeft;
    }
}

```

Bảng 1-2 Nội dung code Form1.cs của Bài 04

```

using System;
using System.Drawing;
using System.Drawing.Text;
using System.Windows.Forms;

namespace Bai04
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
            ImportSystemFonts();
            cbbFont.SelectedItem = "Times New Roman";
            cbbSize.SelectedItem = "16";
            ApplyTextFormat();
        }

        private void ImportSystemFonts()
        {
            InstalledFontCollection installedFont = new InstalledFontCollection();
            FontFamily[] fontFamilies = installedFont.Families;
            foreach (FontFamily family in fontFamilies)
            {
                cbbFont.Items.Add(family.Name);
            }
        }

        private void ApplyTextFormat()
        {
            if (cbbFont.SelectedItem == null || cbbSize.SelectedItem == null)
                return;
            string fontName = cbbFont.SelectedItem.ToString();
            float fontSize = float.Parse(cbbSize.SelectedItem.ToString());
            FontStyle fontStyle = FontStyle.Regular;
            if (chbBold.Checked)
                fontStyle |= FontStyle.Bold;
            if (chbItalic.Checked)
                fontStyle |= FontStyle.Italic;
            if (chbUnderline.Checked)
                fontStyle |= FontStyle.Underline;
            txtText.Font = new Font(fontName, fontSize, fontStyle);
        }

        private void FormatChanged(object sender, EventArgs e)
        {

```

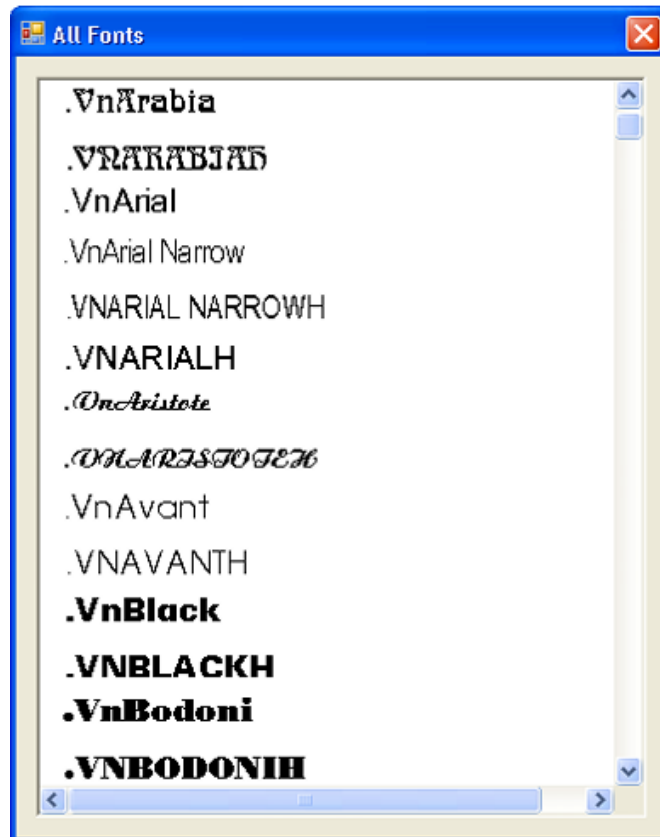
```
        ApplyTextFormat();
    }

    private void btnColor_Click(object sender, EventArgs e)
    {
        ColorDialog colorDialog = new ColorDialog();
        if (colorDialog.ShowDialog() == DialogResult.OK)
        {
            txbText.ForeColor = colorDialog.Color;
            btnColor.BackColor = colorDialog.Color;
        }
    }

    private void AlignChanged(object sender, EventArgs e)
    {
        if (radLeft.Checked)
        {
            txbText.TextAlign = HorizontalAlignment.Left;
        }
        else if (radCenter.Checked)
        {
            txbText.TextAlign = HorizontalAlignment.Center;
        }
        else if (radRight.Checked)
        {
            txbText.TextAlign = HorizontalAlignment.Right;
        }
    }
}
```

Video demo bài 04: [Bài 04 - Font Formatter](#)

Bài 06. Viết chương trình liệt kê tất cả Font chữ có trong hệ thống có dạng như sau:



Bảng 2-1 Nội dung code Form1.Designer.cs của Bài 06

```
namespace Bai06
{
    partial class Form1
    {
        /// <summary>
        /// Required designer variable.
        /// </summary>
        private System.ComponentModel.IContainer components = null;

        /// <summary>
        /// Clean up any resources being used.
        /// </summary>
        /// <param name="disposing">true if managed resources should be disposed;
        otherwise, false.</param>
        protected override void Dispose(bool disposing)
        {
            if (disposing && (components != null))
            {
                components.Dispose();
            }
            base.Dispose(disposing);
        }

        #region Windows Form Designer generated code
```

```

    /// <summary>
    /// Required method for Designer support - do not modify
    /// the contents of this method with the code editor.
    /// </summary>
    private void InitializeComponent()
    {
        this.lsvSystemFonts = new System.Windows.Forms.ListView();
        this.fontCHS = ((System.Windows.Forms.ColumnHeader)(new
System.Windows.Forms.ColumnHeader()));
        this.SuspendLayout();
        //
        // lsvSystemFonts
        //
        this.lsvSystemFonts.Columns.AddRange(new
System.Windows.Forms.ColumnHeader[] {
            this.fontCHS});
        this.lsvSystemFonts.HideSelection = false;
        this.lsvSystemFonts.Location = new System.Drawing.Point(12, 10);
        this.lsvSystemFonts.Name = "lsvSystemFonts";
        this.lsvSystemFonts.Size = new System.Drawing.Size(426, 478);
        this.lsvSystemFonts.TabIndex = 0;
        this.lsvSystemFonts.UseCompatibleStateImageBehavior = false;
        this.lsvSystemFonts.View = System.Windows.Forms.View.Details;
        //
        // fontCHS
        //
        this.fontCHS.Text = "Font";
        this.fontCHS.Width = 450;
        //
        // Form1
        //
        this.AutoScaleDimensions = new System.Drawing.SizeF(8F, 16F);
        this.AutoScaleMode = System.Windows.Forms.AutoScaleMode.Font;
        this.ClientSize = new System.Drawing.Size(450, 500);
        this.Controls.Add(this.lsvSystemFonts);
        this.MaximizeBox = false;
        this.MinimizeBox = false;
        this.Name = "Form1";
        this.Text = "Bài 06 - All Fonts";
        this.ResumeLayout(false);
    }

#endregion

private System.Windows.Forms.ListView lsvSystemFonts;
private System.Windows.Forms.ColumnHeader fontCHS;
}
}

```

Bảng 2-2 Nội dung code Form1.cs của Bài 04

```

using System.Drawing;
using System.Drawing.Text;
using System.Windows.Forms;

namespace Bai06
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
            ImportSystemFonts();
            lsvSystemFonts.HeaderStyle = ColumnHeaderStyle.None;
        }
    }
}

```

```

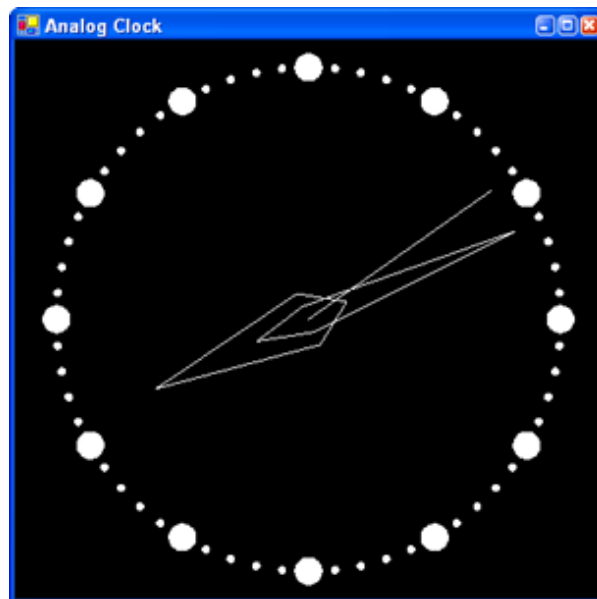
    }

    private void ImportSystemFonts()
    {
        InstalledFontCollection installedFont = new InstalledFontCollection();
        FontFamily[] fontFamilies = installedFont.Families;
        lsvSystemFonts.BeginUpdate();
        foreach (FontFamily family in fontFamilies)
        {
            ListViewItem item = new ListViewItem();
            item.Text = family.Name;
            item.Font = new Font(family, 12);
            lsvSystemFonts.Items.Add(item);
        }
        lsvSystemFonts.EndUpdate();
    }
}
}

```

Video demo bài 06: [Bài 06 - All Fonts](#)

Bài 08: Viết ứng dụng hiển thị giờ hệ thống như hình vẽ sau:



Bảng 3-1 Nội dung code file Form1.Designer.cs của bài 08

```

namespace Bai08
{
    partial class Form1
    {
        /// <summary>
        /// Required designer variable.
        /// </summary>
        private System.ComponentModel.IContainer components = null;

        /// <summary>
        /// Clean up any resources being used.
        /// </summary>
    }
}

```

```

        /// <param name="disposing">true if managed resources should be disposed;
        otherwise, false.</param>
        protected override void Dispose(bool disposing)
        {
            if (disposing && (components != null))
            {
                components.Dispose();
            }
            base.Dispose(disposing);
        }

        #region Windows Form Designer generated code

        /// <summary>
        /// Required method for Designer support - do not modify
        /// the contents of this method with the code editor.
        /// </summary>
        private void InitializeComponent()
        {
            this.components = new System.ComponentModel.Container();
            this.timer1 = new System.Windows.Forms.Timer(this.components);
            this.SuspendLayout();
            //
            // Form1
            //
            this.AutoScaleMode = new System.Drawing.SizeF(8F, 16F);
            this.AutoScaleMode = System.Windows.Forms.AutoScaleMode.Font;
            this.ClientSize = new System.Drawing.Size(420, 420);
            this.BackColor = System.Drawing.Color.Black;
            this.MaximizeBox = false;
            this.MinimizeBox = false;
            this.Name = "Form1";
            this.Text = "Bài 08 - Analog Clock";
            this.ResumeLayout(false);

        }

        #endregion

        private System.Windows.Forms.Timer timer1;
    }
}

```

Bảng 3-2 Nội dung code file Form1.cs của bài 08

```

using System;
using System.Drawing;
using System.Drawing.Drawing2D;
using System.Windows.Forms;

namespace Bai08
{
    public partial class Form1 : Form
    {
        private int _offset = 20;
        public Form1()
        {
            InitializeComponent();
            this.DoubleBuffered = true;
            this.StartPosition = FormStartPosition.CenterScreen;

            timer1.Interval = 10;
            timer1.Tick += TimerClock_Tick;
            timer1.Start();
        }
    }
}

```

```

private void TimerClock_Tick(object sender, EventArgs e)
{
    this.Invalidate();
}

protected override void OnPaint(PaintEventArgs e)
{
    base.OnPaint(e);
    Graphics g = e.Graphics;
    g.SmoothingMode = SmoothingMode.AntiAlias;

    // tâm đồng hồ
    int cx = this.ClientSize.Width / 2;
    int cy = this.ClientSize.Height / 2;
    g.TranslateTransform(cx, cy); // góc tọa độ về tâm đồng hồ

    int radius = Math.Min(cx, cy) - _offset;

    // lưu trạng thái ban đầu của g
    var originalState = g.Save();

    for (int i = 0; i < 60; i++)
    {
        if (i % 5 == 0) // vạch giờ
            g.FillEllipse(Brushes.White, -6, -radius - 5f, 12, 12);
        else // vạch phút
            g.FillEllipse(Brushes.White, -2, -radius, 4, 4);
        g.RotateTransform(6);
    }
    // khôi phục g về trạng thái ban đầu
    g.Restore(originalState);

    // lấy thời gian hiện tại
    DateTime now = DateTime.Now;
    float hour = now.Hour;
    float minute = now.Minute;
    float second = now.Second;

    // vẽ kim giờ: góc = (giờ + phút/60) * 30 độ
    float hourAngle = (hour % 12 + minute / 60f) * 30;

    originalState = g.Save();
    g.RotateTransform(hourAngle);
    PointF[] hourHand = new PointF[]
    {
        new PointF(0, 20),
        new PointF(-10, -radius * 0.2f),
        new PointF(0, -radius * 0.6f),
        new PointF(10, -radius * 0.2f)
    };
    g.DrawPolygon(new Pen(Color.White, 2), hourHand);
    g.Restore(originalState);

    // vẽ kim giờ: góc = (phút + giây/60) * 6 độ
    float minuteAngle = (minute + second / 60f) * 6;
    originalState = g.Save();
    g.RotateTransform(minuteAngle);
    PointF[] minuteHand = new PointF[]
    {
        new PointF(0, 20),
        new PointF(-6, -radius * 0.2f),
        new PointF(0, -radius * 0.9f),
        new PointF(6, -radius * 0.2f)
    };
    g.DrawPolygon(new Pen(Color.White, 2), minuteHand);
    g.Restore(originalState);
}

```

```

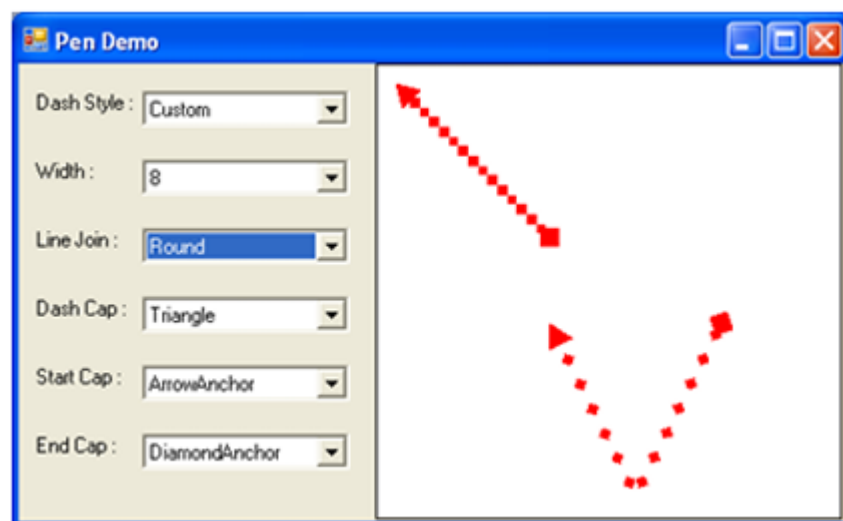
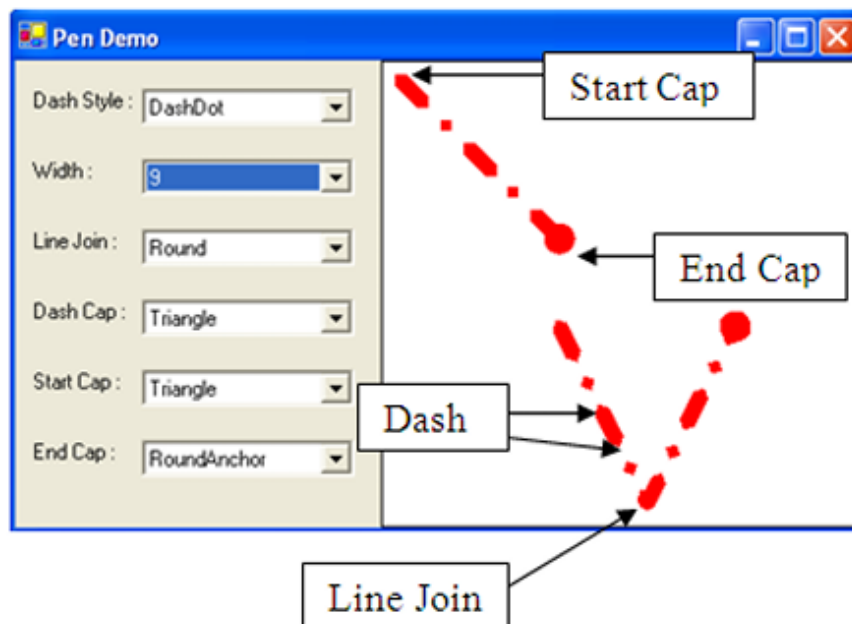
//vẽ kim giây
float secondAngle = (second + now.Millisecond / 1000f) * 6;
originalState = g.Save();
g.RotateTransform(secondAngle);
g.DrawLine(new Pen(Color.Red, 2), 0, 10, 0, -radius * 0.9f);
g.Restore(originalState);

// vẽ tâm đồng hồ
g.FillEllipse(Brushes.White, -5, -5, 10, 10);
    }
}

```

Video demo bài 08: [Bài 08 - Analog Clock](#)

Bài 10: Viết chương trình cho phép demo các loại bút vẽ (pen) như sau:



```

namespace Bai10
{
    partial class Form1
    {
        /// <summary>
        /// Required designer variable.
        /// </summary>
        private System.ComponentModel.IContainer components = null;

        /// <summary>
        /// Clean up any resources being used.
        /// </summary>
        /// <param name="disposing">true if managed resources should be disposed;
        otherwise, false.</param>
        protected override void Dispose(bool disposing)
        {
            if (disposing && (components != null))
            {
                components.Dispose();
            }
            base.Dispose(disposing);
        }

        #region Windows Form Designer generated code

        /// <summary>
        /// Required method for Designer support - do not modify
        /// the contents of this method with the code editor.
        /// </summary>
        private void InitializeComponent()
        {
            this.lbDashStyle = new System.Windows.Forms.Label();
            this.cbDashStyle = new System.Windows.Forms.ComboBox();
            this.pictureBox = new System.Windows.Forms.PictureBox();
            this.cbWidth = new System.Windows.Forms.ComboBox();
            this.lbWidth = new System.Windows.Forms.Label();
            this.cbLineJoin = new System.Windows.Forms.ComboBox();
            this.lbLineJoin = new System.Windows.Forms.Label();
            this.cbDashCap = new System.Windows.Forms.ComboBox();
            this.lbDashCap = new System.Windows.Forms.Label();
            this.cbStartCap = new System.Windows.Forms.ComboBox();
            this.lbStartCap = new System.Windows.Forms.Label();
            this.cbEndCap = new System.Windows.Forms.ComboBox();
            this.lbEndCap = new System.Windows.Forms.Label();

            ((System.ComponentModel.ISupportInitialize)(this.pictureBox)).BeginInit();
            this.SuspendLayout();
            //
            // lbDashStyle
            //
            this.lbDashStyle.AutoSize = true;
            this.lbDashStyle.Location = new System.Drawing.Point(21, 24);
            this.lbDashStyle.Name = "lbDashStyle";
            this.lbDashStyle.Size = new System.Drawing.Size(75, 16);
            this.lbDashStyle.TabIndex = 0;
            this.lbDashStyle.Text = "Dash Style:";
            //
            // cbDashStyle
            //
            this.cbDashStyle.DropDownStyle =
System.Windows.Forms.ComboBoxStyle.DropDownList;
            this.cbDashStyle.FormattingEnabled = true;
            this.cbDashStyle.Items.AddRange(new object[] {
                "Solid",
                "Dash",
                "Dot",
                "DashDot",

```

```

        "DashDotDot"}));
        this.cbDashStyle.Location = new System.Drawing.Point(102, 21);
        this.cbDashStyle.Name = "cbDashStyle";
        this.cbDashStyle.Size = new System.Drawing.Size(157, 24);
        this.cbDashStyle.TabIndex = 1;
        this.cbDashStyle.SelectedIndexChanged += new
System.EventHandler(this.cbDashStyle_SelectedIndexChanged);
        //
        // contentBBox
        //
        this.pictureBox.BackColor = System.Drawing.Color.White;
        this.pictureBox.Location = new System.Drawing.Point(281, 0);
        this.pictureBox.Name = "pictureBox";
        this.pictureBox.Size = new System.Drawing.Size(429, 387);
        this.pictureBox.TabIndex = 2;
        this.pictureBox.TabStop = false;
        this.pictureBox.Paint += new
System.Windows.Forms.PaintEventHandler(this.pictureBox_Paint);
        //
        // cbWidth
        //
        this.cbWidth.DropDownStyle =
System.Windows.Forms.ComboBoxStyle.DropDownList;
        this.cbWidth.FormattingEnabled = true;
        this.cbWidth.Items.AddRange(new object[] {
            "1",
            "2",
            "3",
            "4",
            "5",
            "6",
            "7",
            "8",
            "9",
            "10"});
        this.cbWidth.Location = new System.Drawing.Point(102, 72);
        this.cbWidth.Name = "cbWidth";
        this.cbWidth.Size = new System.Drawing.Size(157, 24);
        this.cbWidth.TabIndex = 4;
        this.cbWidth.SelectedIndexChanged += new
System.EventHandler(this.cbWidth_SelectedIndexChanged);
        //
        // lbWidth
        //
        this.lbWidth.AutoSize = true;
        this.lbWidth.Location = new System.Drawing.Point(21, 75);
        this.lbWidth.Name = "lbWidth";
        this.lbWidth.Size = new System.Drawing.Size(44, 16);
        this.lbWidth.TabIndex = 3;
        this.lbWidth.Text = "Width:";
        //
        // cbLineJoin
        //
        this.cbLineJoin.DropDownStyle =
System.Windows.Forms.ComboBoxStyle.DropDownList;
        this.cbLineJoin.FormattingEnabled = true;
        this.cbLineJoin.Items.AddRange(new object[] {
            "Miter",
            "Bevel",
            "Round",
            "MiterClipped"});
        this.cbLineJoin.Location = new System.Drawing.Point(102, 124);
        this.cbLineJoin.Name = "cbLineJoin";
        this.cbLineJoin.Size = new System.Drawing.Size(157, 24);
        this.cbLineJoin.TabIndex = 6;
        this.cbLineJoin.SelectedIndexChanged += new
System.EventHandler(this.cbLineJoin_SelectedIndexChanged);
        //
        // lbLineJoin

```

```

//
this.lbLineJoin.AutoSize = true;
this.lbLineJoin.Location = new System.Drawing.Point(21, 127);
this.lbLineJoin.Name = "lbLineJoin";
this.lbLineJoin.Size = new System.Drawing.Size(63, 16);
this.lbLineJoin.TabIndex = 5;
this.lbLineJoin.Text = "Line Join:";
//
// cbDashCap
//
this.cbDashCap.DropDownStyle =
System.Windows.Forms.ComboBoxStyle.DropDownList;
this.cbDashCap.FormattingEnabled = true;
this.cbDashCap.Items.AddRange(new object[] {
    "Flat",
    "Round",
    "Triangle"});
this.cbDashCap.Location = new System.Drawing.Point(102, 176);
this.cbDashCap.Name = "cbDashCap";
this.cbDashCap.Size = new System.Drawing.Size(157, 24);
this.cbDashCap.TabIndex = 8;
this.cbDashCap.SelectedIndexChanged += new
System.EventHandler(this.cbDashCap_SelectedIndexChanged);
//
// lbDashCap
//
this.lbDashCap.AutoSize = true;
this.lbDashCap.Location = new System.Drawing.Point(21, 179);
this.lbDashCap.Name = "lbDashCap";
this.lbDashCap.Size = new System.Drawing.Size(70, 16);
this.lbDashCap.TabIndex = 7;
this.lbDashCap.Text = "Dash Cap:";
//
// cbStartCap
//
this.cbStartCap.DropDownStyle =
System.Windows.Forms.ComboBoxStyle.DropDownList;
this.cbStartCap.FormattingEnabled = true;
this.cbStartCap.Items.AddRange(new object[] {
    "Flat",
    "Round",
    "Square",
    "Triangle",
    "ArrowAnchor",
    "RoundAnchor",
    "DiamondAnchor"});
this.cbStartCap.Location = new System.Drawing.Point(102, 228);
this.cbStartCap.Name = "cbStartCap";
this.cbStartCap.Size = new System.Drawing.Size(157, 24);
this.cbStartCap.TabIndex = 10;
this.cbStartCap.SelectedIndexChanged += new
System.EventHandler(this.cbStartCap_SelectedIndexChanged);
//
// lbStartCap
//
this.lbStartCap.AutoSize = true;
this.lbStartCap.Location = new System.Drawing.Point(21, 231);
this.lbStartCap.Name = "lbStartCap";
this.lbStartCap.Size = new System.Drawing.Size(65, 16);
this.lbStartCap.TabIndex = 9;
this.lbStartCap.Text = "Start Cap:";
//
// cbEndCap
//
this.cbEndCap.DropDownStyle =
System.Windows.Forms.ComboBoxStyle.DropDownList;
this.cbEndCap.FormattingEnabled = true;
this.cbEndCap.Items.AddRange(new object[] {
    "Flat",

```

```

        "Round",
        "Square",
        "Triangle",
        "ArrowAnchor",
        "RoundAnchor",
        "DiamondAnchor"});
this.cbEndCap.Location = new System.Drawing.Point(102, 282);
this.cbEndCap.Name = "cbEndCap";
this.cbEndCap.Size = new System.Drawing.Size(157, 24);
this.cbEndCap.TabIndex = 12;

        this.cbEndCap.SelectedIndexChanged += new
            System.EventHandler(this.cbEndCap_SelectedIndexChanged);

//
// lbEndCap
//
this.lbEndCap.AutoSize = true;
this.lbEndCap.Location = new System.Drawing.Point(21, 285);
this.lbEndCap.Name = "lbEndCap";
this.lbEndCap.Size = new System.Drawing.Size(62, 16);
this.lbEndCap.TabIndex = 11;
this.lbEndCap.Text = "End Cap:";
//
// Form1
//
this.AutoScaleDimensions = new System.Drawing.SizeF(8F, 16F);
this.AutoScaleMode = System.Windows.Forms.AutoScaleMode.Font;
this.ClientSize = new System.Drawing.Size(711, 390);
this.Controls.Add(this.cbEndCap);
this.Controls.Add(this.lbEndCap);
this.Controls.Add(this.cbStartCap);
this.Controls.Add(this.lbStartCap);
this.Controls.Add(this.cbDashCap);
this.Controls.Add(this.lbDashCap);
this.Controls.Add(this.cbLineJoin);
this.Controls.Add(this.lbLineJoin);
this.Controls.Add(this.cbWidth);
this.Controls.Add(this.lbWidth);
this.Controls.Add(this.pictureBox);
this.Controls.Add(this.cbDashStyle);
this.Controls.Add(this.lbDashStyle);
this.MaximizeBox = false;
this.MinimizeBox = false;
this.Name = "Form1";
this.Text = "Pen Demo";

((System.ComponentModel.ISupportInitialize)(this.pictureBox)).EndInit();
this.ResumeLayout(false);
this.PerformLayout();

}

#endregion

private System.Windows.Forms.Label lbDashStyle;
private System.Windows.Forms.ComboBox cbDashStyle;
private System.Windows.Forms.PictureBox contentBox;
private System.Windows.Forms.ComboBox cbWidth;
private System.Windows.Forms.Label lbWidth;
private System.Windows.Forms.ComboBox cbLineJoin;
private System.Windows.Forms.Label lbLineJoin;
private System.Windows.Forms.ComboBox cbDashCap;
private System.Windows.Forms.Label lbDashCap;
private System.Windows.Forms.ComboBox cbStartCap;
private System.Windows.Forms.Label lbStartCap;
private System.Windows.Forms.ComboBox cbEndCap;
private System.Windows.Forms.Label lbEndCap;
}

```

}

Bảng 4-2 Nội dung code Form1.cs của bài 10

```

using System;
using System.Drawing;
using System.Windows.Forms;

namespace Bai10
{
    public partial class Form1 : Form
    {
        Bitmap bitmap;
        Graphics bitmapGraphics;
        private Pen _mainPen = new Pen(Color.Red, 1);
        public Form1()
        {
            InitializeComponent();
            bitmap = new Bitmap(contentBox.Width, contentBox.Height);
            bitmapGraphics = Graphics.FromImage(bitmap);
            contentBox.Image = bitmap;

            float[] dashPattern = { 1f, 5f };
            _mainPen.DashPattern = dashPattern;

            SetComboBoxItems();
        }

        private void SetComboBoxItems()
        {
            cbDashStyle.SelectedIndex = 0;
            cbWidth.SelectedIndex = 0;
            cbLineJoin.SelectedIndex = 0;
            cbDashCap.SelectedIndex = 0;
            cbStartCap.SelectedIndex = 0;
            cbEndCap.SelectedIndex = 0;
        }

        private void DrawLine(object sender, PaintEventArgs e)
        {
            bitmapGraphics.Clear(Color.White);

            Point p1 = new Point(30, 30);
            Point p2 = new Point(150, 150);
            bitmapGraphics.DrawLine(_mainPen, p1, p2);

            Point[] polyPoints = new Point[]
            {
                new Point(30, 180),
                new Point(150, 300),
                new Point(200, 180)
            };
            bitmapGraphics.DrawLines(_mainPen, polyPoints);
            contentBox.Refresh();
        }

        private void cbDashStyle_SelectedIndexChanged(object sender, EventArgs e)
        {
            if (cbDashStyle.SelectedIndex > -1)
            {
                string selectedValue = cbDashStyle.SelectedItem.ToString();
                _mainPen.DashStyle =
                (System.Drawing.Drawing2D.DashStyle)Enum.Parse(
                    typeof(System.Drawing.Drawing2D.DashStyle),
                    selectedValue);
            }
        }
    }
}

```

```

    }
}

private void cbWidth_SelectedIndexChanged(object sender, EventArgs e)
{
    if (cbWidth.SelectedIndex > -1)
    {
        if (float.TryParse(cbWidth.SelectedItem.ToString(), out float
width))
        {
            _mainPen.Width = width;
        }
    }
}

private void cbLineJoin_SelectedIndexChanged(object sender, EventArgs e)
{
    if (cbLineJoin.SelectedIndex > -1)
    {
        _mainPen.LineJoin = (System.Drawing.Drawing2D.LineJoin)Enum.Parse(
typeof(System.Drawing.Drawing2D.LineJoin),
cbLineJoin.SelectedItem.ToString());
    }
}

private void cbDashCap_SelectedIndexChanged(object sender, EventArgs e)
{
    if (cbDashCap.SelectedIndex > -1)
    {
        _mainPen.DashCap = (System.Drawing.Drawing2D.DashCap)Enum.Parse(
typeof(System.Drawing.Drawing2D.DashCap),
cbDashCap.SelectedItem.ToString());
    }
}

private void cbStartCap_SelectedIndexChanged(object sender, EventArgs e)
{
    if (cbStartCap.SelectedIndex > -1)
    {
        _mainPen.StartCap = (System.Drawing.Drawing2D.LineCap)Enum.Parse(
typeof(System.Drawing.Drawing2D.LineCap),
cbStartCap.SelectedItem.ToString());
    }
}

private void cbEndCap_SelectedIndexChanged(object sender, EventArgs e)
{
    if (cbEndCap.SelectedIndex > -1)
    {
        _mainPen.EndCap = (System.Drawing.Drawing2D.LineCap)Enum.Parse(
typeof(System.Drawing.Drawing2D.LineCap),
cbEndCap.SelectedItem.ToString());
    }
}
}
}

```

Video demo bài 10: [Bài 10 - Pen Demo](#)

Bài 11: Xây dựng chương trình vẽ hình cơ bản như sau:

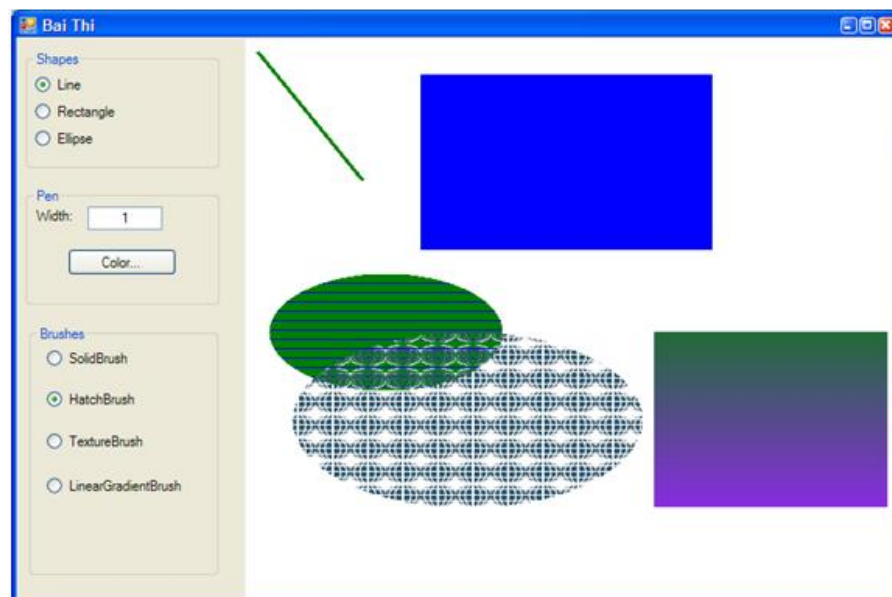
Khi chọn button “Color” thì xuất hiện hộp thoại Color để chọn màu vẽ đường.

Khi người dùng nhấn giữ nút trái chuột và di chuyển chuột trên pictureBox thì sẽ vẽ hình tương ứng tùy thuộc vào hình được chọn:

- Line: Vẽ đường thẳng có độ dày (txtWidth) và màu tương ứng đã được chọn.

- Rectangle hay Ellipse: Tô hình chữ nhật hay hình Ellipse với Brush tương ứng:

- SolidBrush: sử dụng SolidBrush với màu là Green.
- HatchBrush: sử dụng HatchBrush có kiểu là Horizontal với 2 màu tương ứng là Blue và Green
- TextureBrush: sử dụng TextureBrush, tô với bitmap cho trước.
- LinearGradientBrush: sử dụng LinearGradientBrush có kiểu là Vertical với 2 màu tương ứng là: Red và Green.



```

namespace Bai11
{
    partial class Form1
    {
        /// <summary>
        /// Required designer variable.
        /// </summary>
        private System.ComponentModel.IContainer components = null;

        /// <summary>
        /// Clean up any resources being used.
        /// </summary>
        /// <param name="disposing">true if managed resources should be disposed;
        otherwise, false.</param>
        protected override void Dispose(bool disposing)
        {
            if (disposing && (components != null))
            {
                components.Dispose();
            }
            base.Dispose(disposing);
        }

        #region Windows Form Designer generated code

        /// <summary>
        /// Required method for Designer support - do not modify
        /// the contents of this method with the code editor.
        /// </summary>
        private void InitializeComponent()
        {
            this.contentBox = new System.Windows.Forms.PictureBox();
            this.panel1 = new System.Windows.Forms.Panel();
            this.groupBox3 = new System.Windows.Forms.GroupBox();
            this.rabLinearGradientBRush = new System.Windows.Forms.RadioButton();
            this.rabTextureBrush = new System.Windows.Forms.RadioButton();
            this.rabHatchBrush = new System.Windows.Forms.RadioButton();
            this.rabSolidBrush = new System.Windows.Forms.RadioButton();
            this.groupBox2 = new System.Windows.Forms.GroupBox();
            this.label2 = new System.Windows.Forms.Label();
            this.btnColor = new System.Windows.Forms.Button();
            this.label1 = new System.Windows.Forms.Label();
            this.cbbWidth = new System.Windows.Forms.ComboBox();
            this.groupBox1 = new System.Windows.Forms.GroupBox();
            this.rabEllipse = new System.Windows.Forms.RadioButton();
            this.rabRectangle = new System.Windows.Forms.RadioButton();
            this.rabLine = new System.Windows.Forms.RadioButton();

            ((System.ComponentModel.ISupportInitialize)(this.contentBox)).BeginInit();
            this.panel1.SuspendLayout();
            this.groupBox3.SuspendLayout();
            this.groupBox2.SuspendLayout();
            this.groupBox1.SuspendLayout();
            this.SuspendLayout();
            //
            // contentBox
            //
            this.contentBox.BackColor = System.Drawing.Color.White;
            this.contentBox.Dock = System.Windows.Forms.DockStyle.Fill;
            this.contentBox.Location = new System.Drawing.Point(0, 0);
            this.contentBox.Name = "contentBox";
            this.contentBox.Size = new System.Drawing.Size(800, 450);
            this.contentBox.TabIndex = 1;
            this.contentBox.TabStop = false;
            this.contentBox.MouseDown += new
            System.Windows.Forms.MouseEventHandler(this.pictureBox_MouseDown);
        }
    }
}

```

```

        this.contentBox.MouseUp += new
System.Windows.Forms.MouseEventHandler(this.pictureBox_MouseUp);
        //
        // panel1
        //
        this.panel1.Controls.Add(this.groupBox3);
        this.panel1.Controls.Add(this.groupBox2);
        this.panel1.Controls.Add(this.groupBox1);
        this.panel1.Dock = System.Windows.Forms.DockStyle.Left;
        this.panel1.Location = new System.Drawing.Point(0, 0);
        this.panel1.Name = "panel1";
        this.panel1.Size = new System.Drawing.Size(197, 450);
        this.panel1.TabIndex = 2;
        //
        // groupBox3
        //
        this.groupBox3.BackColor = System.Drawing.Color.White;
        this.groupBox3.Controls.Add(this.rabLinearGradientBRush);
        this.groupBox3.Controls.Add(this.rabTextureBrush);
        this.groupBox3.Controls.Add(this.rabHatchBrush);
        this.groupBox3.Controls.Add(this.rabSolidBrush);
        this.groupBox3.Location = new System.Drawing.Point(8, 279);
        this.groupBox3.Name = "groupBox3";
        this.groupBox3.Size = new System.Drawing.Size(178, 159);
        this.groupBox3.TabIndex = 3;
        this.groupBox3.TabStop = false;
        this.groupBox3.Text = "Brushes";
        //
        // rabLinearGradientBRush
        //
        this.rabLinearGradientBRush.AutoSize = true;
        this.rabLinearGradientBRush.Location = new System.Drawing.Point(6,
124);
        this.rabLinearGradientBRush.Name = "rabLinearGradientBRush";
        this.rabLinearGradientBRush.Size = new System.Drawing.Size(150, 20);
        this.rabLinearGradientBRush.TabIndex = 3;
        this.rabLinearGradientBRush.TabStop = true;
        this.rabLinearGradientBRush.Text = "LinearGradientBrush";
        this.rabLinearGradientBRush.UseVisualStyleBackColor = true;
        this.rabLinearGradientBRush.CheckedChanged += new
System.EventHandler(this.BrushChanged);
        //
        // rabTextureBrush
        //
        this.rabTextureBrush.AutoSize = true;
        this.rabTextureBrush.Location = new System.Drawing.Point(6, 92);
        this.rabTextureBrush.Name = "rabTextureBrush";
        this.rabTextureBrush.Size = new System.Drawing.Size(107, 20);
        this.rabTextureBrush.TabIndex = 2;
        this.rabTextureBrush.TabStop = true;
        this.rabTextureBrush.Text = "TextureBrush";
        this.rabTextureBrush.UseVisualStyleBackColor = true;
        this.rabTextureBrush.CheckedChanged += new
System.EventHandler(this.BrushChanged);
        //
        // rabHatchBrush
        //
        this.rabHatchBrush.AutoSize = true;
        this.rabHatchBrush.Location = new System.Drawing.Point(6, 60);
        this.rabHatchBrush.Name = "rabHatchBrush";
        this.rabHatchBrush.Size = new System.Drawing.Size(97, 20);
        this.rabHatchBrush.TabIndex = 1;
        this.rabHatchBrush.TabStop = true;
        this.rabHatchBrush.Text = "HatchBrush";
        this.rabHatchBrush.UseVisualStyleBackColor = true;
        this.rabHatchBrush.CheckedChanged += new
System.EventHandler(this.BrushChanged);
        //
        // rabSolidBrush

```

```

//
this.rabSolidBrush.AutoSize = true;
this.rabSolidBrush.Location = new System.Drawing.Point(6, 30);
this.rabSolidBrush.Name = "rabSolidBrush";
this.rabSolidBrush.Size = new System.Drawing.Size(93, 20);
this.rabSolidBrush.TabIndex = 0;
this.rabSolidBrush.TabStop = true;
this.rabSolidBrush.Text = "SolidBrush";
this.rabSolidBrush.UseVisualStyleBackColor = true;
this.rabSolidBrush.CheckedChanged += new
System.EventHandler(this.BrushChanged);
//
// groupBox2
//
this.groupBox2.BackColor = System.Drawing.Color.White;
this.groupBox2.Controls.Add(this.label2);
this.groupBox2.Controls.Add(this.btnColor);
this.groupBox2.Controls.Add(this.label1);
this.groupBox2.Controls.Add(this.cbbWidth);
this.groupBox2.Location = new System.Drawing.Point(8, 162);
this.groupBox2.Name = "groupBox2";
this.groupBox2.Size = new System.Drawing.Size(177, 101);
this.groupBox2.TabIndex = 1;
this.groupBox2.TabStop = false;
this.groupBox2.Text = "Pen";
//
// label2
//
this.label2.AutoSize = true;
this.label2.Location = new System.Drawing.Point(21, 72);
this.label2.Name = "label2";
this.label2.Size = new System.Drawing.Size(39, 16);
this.label2.TabIndex = 3;
this.label2.Text = "Color";
//
// btnColor
//
this.btnColor.BackColor = System.Drawing.Color.White;
this.btnColor.FlatStyle = System.Windows.Forms.FlatStyle.Popup;
this.btnColor.Location = new System.Drawing.Point(71, 69);
this.btnColor.Name = "btnColor";
this.btnColor.Size = new System.Drawing.Size(84, 22);
this.btnColor.TabIndex = 2;
this.btnColor.UseVisualStyleBackColor = false;
this.btnColor.Click += new System.EventHandler(this.btnColor_Click_1);
//
// label1
//
this.label1.AutoSize = true;
this.label1.Location = new System.Drawing.Point(21, 33);
this.label1.Name = "label1";
this.label1.Size = new System.Drawing.Size(41, 16);
this.label1.TabIndex = 1;
this.label1.Text = "Width";
//
// cbbWidth
//
this.cbbWidth.FormattingEnabled = true;
this.cbbWidth.Items.AddRange(new object[] {
    "1",
    "2",
    "3",
    "4",
    "5",
    "6",
    "7",
    "8",
    "9",
    "10"});

```

```

        this.cbbWidth.Location = new System.Drawing.Point(71, 30);
        this.cbbWidth.Name = "cbbWidth";
        this.cbbWidth.Size = new System.Drawing.Size(84, 24);
        this.cbbWidth.TabIndex = 0;
        this.cbbWidth.SelectedIndexChanged += new
            System.EventHandler(this.cbbWidth_SelectedIndexChanged);
    //
    // groupBox1
    //
    this.groupBox1.BackColor = System.Drawing.Color.White;
    this.groupBox1.Controls.Add(this.rabEllipse);
    this.groupBox1.Controls.Add(this.rabRectangle);
    this.groupBox1.Controls.Add(this.rabLine);
    this.groupBox1.Location = new System.Drawing.Point(8, 18);
    this.groupBox1.Name = "groupBox1";
    this.groupBox1.Size = new System.Drawing.Size(178, 120);
    this.groupBox1.TabIndex = 0;
    this.groupBox1.TabStop = false;
    this.groupBox1.Text = "Shapes";
    //
    // rabEllipse
    //
    this.rabEllipse.AutoSize = true;
    this.rabEllipse.Location = new System.Drawing.Point(6, 83);
    this.rabEllipse.Name = "rabEllipse";
    this.rabEllipse.Size = new System.Drawing.Size(69, 20);
    this.rabEllipse.TabIndex = 2;
    this.rabEllipse.TabStop = true;
    this.rabEllipse.Text = "Ellipse";
    this.rabEllipse.UseVisualStyleBackColor = true;
    //
    // rabRectangle
    //
    this.rabRectangle.AutoSize = true;
    this.rabRectangle.Location = new System.Drawing.Point(6, 57);
    this.rabRectangle.Name = "rabRectangle";
    this.rabRectangle.Size = new System.Drawing.Size(90, 20);
    this.rabRectangle.TabIndex = 1;
    this.rabRectangle.TabStop = true;
    this.rabRectangle.Text = "Rectangle";
    this.rabRectangle.UseVisualStyleBackColor = true;
    //
    // rabLine
    //
    this.rabLine.AutoSize = true;
    this.rabLine.Location = new System.Drawing.Point(6, 31);
    this.rabLine.Name = "rabLine";
    this.rabLine.Size = new System.Drawing.Size(53, 20);
    this.rabLine.TabIndex = 0;
    this.rabLine.TabStop = true;
    this.rabLine.Text = "Line";
    this.rabLine.UseVisualStyleBackColor = true;
    //
    // Form1
    //
    this.AutoScaleDimensions = new System.Drawing.SizeF(8F, 16F);
    this.AutoScaleModeMode = System.Windows.Forms.AutoScaleModeMode.Font;
    this.ClientSize = new System.Drawing.Size(800, 450);
    this.Controls.Add(this.panel1);
    this.Controls.Add(this.contentBox);
    this.Name = "Form1";
    this.Text = "Form1";

    ((System.ComponentModel.ISupportInitialize)(this.contentBox)).EndInit();
    this.panel1.ResumeLayout(false);
    this.groupBox3.ResumeLayout(false);
    this.groupBox3.PerformLayout();
    this.groupBox2.ResumeLayout(false);
    this.groupBox2.PerformLayout();

```

```

        this.groupBox1.ResumeLayout(false);
        this.groupBox1.PerformLayout();
        this.ResumeLayout(false);

    }

    #endregion
    private System.Windows.Forms.PictureBox contentBox;
    private System.Windows.Forms.Panel panel1;
    private System.Windows.Forms.GroupBox groupBox1;
    private System.Windows.Forms.RadioButton rabEllipse;
    private System.Windows.Forms.RadioButton rabRectangle;
    private System.Windows.Forms.RadioButton rabLine;
    private System.Windows.Forms.GroupBox groupBox3;
    private System.Windows.Forms.RadioButton rabLinearGradientBrush;
    private System.Windows.Forms.RadioButton rabTextureBrush;
    private System.Windows.Forms.RadioButton rabHatchBrush;
    private System.Windows.Forms.RadioButton rabSolidBrush;
    private System.Windows.Forms.GroupBox groupBox2;
    private System.Windows.Forms.Label label2;
    private System.Windows.Forms.Button btnColor;
    private System.Windows.Forms.Label label1;
    private System.Windows.Forms.ComboBox cbbWidth;
}

```

Bảng 5-2 Nội dung code Form1.cs của bài 11

```

using System;
using System.Drawing;
using System.Windows.Forms;

namespace Bai11
{
    public partial class Form1 : Form
    {
        ColorDialog colorDialog = new ColorDialog();
        Bitmap bitmap;
        Graphics bitmapGraphics;
        Pen pen = new Pen(Color.Red);
        Point lastPos;
        public Form1()
        {
            InitializeComponent();
            bitmap = new Bitmap(contentBox.Width, contentBox.Height);
            bitmapGraphics = Graphics.FromImage(bitmap);
            bitmapGraphics.Clear(Color.White);
            contentBox.Image = bitmap;

            cbbWidth.SelectedItem = "4";
            btnColor.BackColor = Color.Red;
            rabSolidBrush.Checked = true;
            rabLine.Checked = true;
        }

        private void btnColor_Click(object sender, EventArgs e)
        {
            if (colorDialog.ShowDialog() == DialogResult.OK)
            {
                btnColor.BackColor = colorDialog.Color;
                pen.Color = colorDialog.Color;
            }
        }

        private void pictureBox_MouseDown(object sender, MouseEventArgs e)
        {
            lastPos = e.Location;
        }
    }
}

```

```

    }

    //private void pictureBox_MouseMove(object sender, MouseEventArgs e)
    //{
    //    if (rabLine.Checked)
    //    {
    //        if (e.Button == MouseButtons.Left)
    //        {
    //            bitmapGraphics.DrawLine(pen, lastPos, e.Location);
    //            lastPos = e.Location;
    //            contentBox.Refresh();
    //        }
    //    }
    //}

    private void pictureBox_MouseUp(object sender, MouseEventArgs e)
    {
        if(rabLine.Checked)
        {
            if (e.Button == MouseButtons.Left)
            {
                bitmapGraphics.DrawLine(pen, lastPos, e.Location);
                contentBox.Refresh();
            }
        }
        else if (rabRectangle.Checked)
        {
            if (e.Button == MouseButtons.Left)
            {
                int x = Math.Min(lastPos.X, e.X);
                int y = Math.Min(lastPos.Y, e.Y);
                int width = Math.Abs(e.X - lastPos.X);
                int height = Math.Abs(e.Y - lastPos.Y);
                bitmapGraphics.FillRectangle(pen.Brush, x, y, width, height);
                contentBox.Refresh();
            }
        }
        else if (rabEllipse.Checked)
        {
            if (e.Button == MouseButtons.Left)
            {
                int x = Math.Min(lastPos.X, e.X);
                int y = Math.Min(lastPos.Y, e.Y);
                int width = Math.Abs(e.X - lastPos.X);
                int height = Math.Abs(e.Y - lastPos.Y);
                bitmapGraphics.FillEllipse(pen.Brush, x, y, width, height);
                contentBox.Refresh();
            }
        }
    }

    private void cbbWidth_SelectedIndexChanged(object sender, EventArgs e)
    {
        pen.Width = float.Parse(cbbWidth.SelectedItem.ToString());
    }

    private void BrushChanged(object sender, EventArgs e)
    {
        if (rabSolidBrush.Checked)
        {
            pen.Brush = new SolidBrush(pen.Color);
        }
        else if (rabHatchBrush.Checked)
        {
            pen.Brush = new System.Drawing.Drawing2D.HatchBrush(
                System.Drawing.Drawing2D.HatchStyle.Horizontal,
                Color.Blue,
                Color.Green);
        }
    }

```

```

        else if (rabTextureBrush.Checked)
        {
            int textureSize = 16;
            Bitmap textureBitmap = new Bitmap(textureSize, textureSize);
            using (Graphics g = Graphics.FromImage(textureBitmap))
            {
                g.Clear(Color.LightGray);
                SolidBrush dotBrush = new SolidBrush(Color.Black);
                g.FillEllipse(dotBrush,
                    textureSize / 4,
                    textureSize / 4,
                    textureSize / 2,
                    textureSize / 2);
            }
            pen.Brush = new TextureBrush(textureBitmap);
        }
        else if (rabLinearGradientBrush.Checked)
        {
            pen.Brush = new System.Drawing.Drawing2D.LinearGradientBrush(
                new Point(0, 0),
                new Point(0, 200),
                Color.Red,
                Color.Green);
        }
    }

    private void btnColor_Click_1(object sender, EventArgs e)
    {
        ColorDialog colorDialog = new ColorDialog();
        if (colorDialog.ShowDialog() == DialogResult.OK)
        {
            btnColor.BackColor = colorDialog.Color;
            pen.Color = colorDialog.Color;
        }
    }
}

```

Video demo bài 11: [Bài 11 - Shapes & Brushes](#)

Github: [BTH5 NguyenChiNguyen 24521186](#)

Drive video: [BTH5 NguyenChiNguyen 24521186](#)